Ruipu Shi

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ABOUT ME

Data Analyst with a background in computing and hands-on experience working with complex real-world datasets across government, geospatial and business domains.
Skilled in Python, SQL, Power BI and data visualisation, with experience in multi-source data integration, data quality analysis, statistical and machine-learning evaluation, and translating complex datasets into clear, decision-ready insights.

KEY STRENGTHS

Team Collaboration & Leadership

Analytical & Problem-Solving Mindset

Communication & Stakeholder Engagement

Adaptability & Initiative

Attention to Detail & Quality Assurance

WORK EXPERIENCE

Urban Intelligence, Canberra
Aug 2026 - Present

Junior Data Analyst Intern

- Investigated and integrated undocumented ACT development application datasets across multiple nested JSON and GeoJSON sources, reconstructing relationships between datasets and normalising data for analysis and internal use.
- Developed geospatial analysis workflows using latitude/longitude, point and polygon geometries to associate government land and development records with internally defined parcel polygons and identifiers.
- Analysed data coverage, structure and cross-source consistency to identify missing records, duplicated entities and differences between government datasets.
- Built a document lifecycle monitoring framework for nearly 300 external documents, implementing structured source/version tracking, automated change detection and rule-based successor identification with manual review for ambiguous cases.

QX Tech, Canberra
Nov 2025 - Present

Data Engineer Intern

- Analysed large-scale company datasets using Python and SQL, identifying missing, duplicated and inconsistent records and improving data completeness and reliability.
- Built automated data collection and enrichment workflows using web and API sources to transform unstructured company information into structured datasets for downstream analysis and applications.
- Developed data-quality and entity-resolution workflows combining deterministic validation rules with automated extraction, enrichment and deduplication.
- Applied text classification models to business categorisation tasks and used confidence-based thresholds to separate reliable predictions from cases requiring manual review.

EDUCATION

Master of Data Science
University of Canberra
July 2025 - Present

Bachelor of Computing
Australian National University
July 2020 - Dec 2024

TECHNICAL SKILLS

Data Analysis	Python (Pandas, NumPy), SQL, R, Exploratory Data Analysis, Data Cleaning, Data Validation
Statistics & Analytics	Statistical Analysis, Classification, Scikit-learn, PyTorch, Feature Engineering, Model Evaluation, Data Preprocessing
Data Visualisation	Dashboard Design, Power BI, Tableau, Excel, Plotly, Data Storytelling
Tools & Platforms	Git, Linux, AWS Fundamentals (EC2, S3, VPC)

ACADEMIC & TECHNICAL PROJECTS

Physiological Signal Classification in VR | ML Research Project
Jun 2026 – Aug 2026

- Replicated and extended a prior conference study using physiological data from several participants, analysing BVP, EDA and respiration signals recorded across baseline and three VR activity conditions.
- Applied a shared preprocessing and feature-extraction pipeline to derive 51 physiological features across the three sensor modalities.
- Developed three-class activity classifiers using KNN, SVM and Random Forest and evaluated performance using accuracy and macro-F1.
- Applied subject-level validation with a held-out participant and nested Leave-One-Subject-Out Validation for feature selection and model hyperparameter optimisation.
- Compared machine-learning feature selection with independent statistical analysis, identifying substantial overlap in the selected physiological features.

Executive Analytics Dashboard | ANU Data Transformation Initiative
Feb 2024 – Nov 2024

- Collaborated with stakeholders to identify reporting needs and define key business metrics.
- Analysed organisational data and developed interactive dashboards using Power BI and Python.
- Translated complex datasets into clear visual insights to support executive decision-making.
- Delivered a user-centred analytics solution that received a Distinction grade.

Pet Image Classification System | Computer Vision
Mar 2026

- Built an image classification pipeline using PyTorch, fine-tuning a pre-trained ResNet18 model for binary classification
- Applied data preprocessing and augmentation techniques (e.g., normalization, transformations) to improve model generalisation
- Evaluated model performance using validation metrics and confusion matrix, achieving ~94% accuracy on unseen data

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References Available Upon Request